

evidence that during folding an early framework intermediate forms in which there is significant secondary structure^{2,3} and the proposal that the transition state occurs late on the folding pathway¹. The initial results from our protein engineering study have begun to map out the energetics of these processes and to show new details such as a major loop not being formed in the

transition state or the lack of hydrogen bonds from N caps to the exposed backbone NH groups of the helices. These studies now point to new questions that can be answered by studying further mutations. For example, are all the loops exposed to solvent in the transition state and are they the last elements of structure to be formed? □

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LETTERS TO NATURE

Nucleosynthesis, neutrino bursts and γ -rays from coalescing neutron stars

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NEUTRON-STAR collisions occur inevitably when binary neutron stars spiral into each other as a result of damping of gravitational radiation. Such collisions will produce a characteristic burst of gravitational radiation, which may be the most promising source of a detectable signal for proposed gravity-wave detectors¹. Such signals are sufficiently unique and robust for them to have been proposed as a means of determining the Hubble constant². However, the rate of these neutron-star collisions is highly uncertain³. Here we note that such events should also synthesize neutron-rich heavy elements, thought to be formed by rapid neutron capture (the r-process)⁴. Furthermore, these collisions should produce neutrino bursts⁵ and resultant bursts of γ -rays; the latter should comprise a subclass of observable γ -ray bursts. We argue that observed r-process abundances and γ -ray-burst rates predict rates for these collisions that are both significant and consistent with other estimates.

The binary pulsar system will coalesce in roughly 10^8 yr. Using this fact together with the pulsar birth rate and the observed frequency of occurrence of close binary pulsars (then, one in every 300 pulsars), Clark *et al.*³ estimated the formation rate in the Galaxy to be $(3 \pm 1.6) \times 10^{-4} \text{ yr}^{-1}$. As 450 pulsars have now been observed, this estimate should be decreased by a factor of 1.5, but this does not change the following discussion. The natural progenitor for a close neutron-star binary system is a massive X-ray binary, from which the neutron-star binary will form following a supernova explosion. Clark *et al.*³ estimated that the probability of disruption of such a close binary in the

supernova of the secondary is 0.1–0.2, and this led them to a formation rate consistent with that based on pulsar statistics.

Alternatively, neutron stars formed in separate events might become associated in a dense globular cluster^{6,7}. Such modes of formation will lead to roughly equal-mass binaries with both neutron stars of mass $1.4 M_{\odot}$.

On the other hand, were there to exist a separate class of neutron-star binaries in which the stars are in close proximity on formation of the binary, these might escape detection because they would decay on a short timescale, τ_{gr} , via gravitational radiation. For example, it has been suggested recently⁶ that millisecond pulsars arise from accretion-induced collapse of white dwarfs in binaries⁷. It is argued on statistical grounds⁶ that low-mass X-ray binaries are too rare to be the progenitors of millisecond pulsars, as had been proposed previously⁸. As the white dwarfs are likely to be spun up to maximum angular momentum, the collapse to a neutron star is likely to 'fizzle', leading to two neutron stars in orbit around each other⁹. It is not unlikely that the two neutron stars would have significantly different masses. Although the statistical arguments have been criticized (F. Verbunt, J. van Paradijs and W. Lewin, personal communication), such criticism does not exclude the possibility that close neutron-star binaries are indeed formed from white dwarfs in globular clusters.

Essentially for all equations of state, the less massive component (the secondary) has a larger radius¹⁰, R_2 . As it also has the smaller Roche-lobe radius, R_L , it is invariably this lighter component that fills its tidal lobe first, following which mass transfer ensues. If $\zeta_2 \equiv (\partial \ln R_2 / \partial \ln M_2) > \zeta_L \equiv (\partial \ln R_L / \partial \ln M_2)$ the mass transfer is dynamically stable, whereas for $\zeta_2 < \zeta_L$ it is unstable. The latter situation arises when the secondary ('donor') neutron star has a mass that is comparable to or only slightly smaller than that of the larger, whereas a high initial mass ratio will result in a stable mass-stripping process.

Clark and Eardley¹ have discussed the evolution of a close neutron-star binary. In systems that are unstable to mass transfer on a dynamical timescale, the margin by which the less massive neutron star is the first to overfill its lobe grows until the orbital evolution is dominated by mass exchange (rather than by gravitational radiation), for which the timescale¹¹ Δt_{ex} is ~ 6 ms for masses of $M_1 = 1.4 M_{\odot}$, $M_2 = 1.2 M_{\odot}$. Once the mass-transfer timescale becomes shorter than the gravitational-radiation timescale, mass loss accelerates rapidly. In a three-dimensional simulation of this process for a doubly degenerate binary system, it is found that the lighter component is completely dissipated in a little more than two orbital periods (~ 4 ms in our case).

The secondary is transformed into a thick, axially symmetric disk orbiting around the primary. Such configurations for the case of a white-dwarf binary have been constructed recently¹². In the calculation of Benz *et al.*¹³, about 0.3% of the total mass escaped the system. A similar fraction can be expected in our case, because this fraction depends mainly on the surface potentials of the two stars and is thus proportional to $(1 - M_2 R_1 / M_1 R_2)$.

Calculations^{12,13} have shown that, for the white-dwarf binary, the collapse does not occur immediately even though the total mass exceeds the Chandrasekhar mass, mainly because the system is supported by centrifugal forces (stability is also indicated by the ratio of kinetic to gravitational energy). Thus, the late-stage evolution of the configuration depends critically on transport and disposal of angular momentum from the disk. If transport of angular momentum occurs entirely by means of degenerate matter viscosity¹⁴, then extremely long viscous timescales result. The value of the Reynolds number indicates, however, that turbulence may arise in a significant fraction of the disk¹². In this case, the relevant viscosity may be the turbulent viscosity

$$\nu_{\text{turb}} \approx (2 \times 10^9 \text{ cm}^2 \text{ s}^{-1}) (V_t / 10^8 \text{ cm s}^{-1}) (l_t / 10^5 \text{ cm}) (\text{Re}_c / 5,000)^{-1},$$

where l_t and V_t are the typical size and velocity of a turbulent cell and Re_c is the critical Reynolds number. Such a turbulent disk will transport angular momentum on a timescale

$$\tau_{\text{vis}} \approx 500 (R / 10^6 \text{ cm})^2 (\nu_{\text{turb}} / 2 \times 10^9 \text{ cm}^2 \text{ s}^{-1})^{-1} \text{ s}.$$

If the binary is stable ($\zeta_2 > \zeta_1$) against dynamical-timescale mass transfer (that is, if the initial mass ratio is large), the mass-transfer rate, $\dot{M}_2$, will approach asymptotically a value dictated essentially by gravitational radiation¹⁵: $\dot{M}_2 \approx -2M_2 / (\zeta_2 - \zeta_1) \tau_{\text{gr}}$. This accretion rate is enormous and it raises again the question of how angular momentum is disposed of and of whether the accreting neutron star has the ability to accommodate such a large mass influx. In the case of stable mass transfer there is another interesting possibility. Neutron stars have a minimum mass below which they are unstable to free expansion. In the present context, the secondary may be stripped to this minimum mass before colliding with the more massive component, at which stage it will explode¹⁶⁻¹⁸.

An interesting feature of the decompression of the secondary is the accompanying de-neutronization of the material. One expects the neutrons to aggregate into large droplets, in the same way as they do in decompressed water^{19,20}. Assuming that the difference in Fermi energies of the neutron and proton in the neutron-rich nuclei is of the order of a few MeV, the β -decay timescale is of the order of a few milliseconds near the neutron drip line (where neutron-rich nuclei are no longer able to capture or retain additional neutrons), which is comparable to the decompression timescale. Hence the de-neutronization is non-adiabatic. This has two important implications. First, decompression can occur while the nuclei are far from the β -decay instability, and thus the nuclei may undergo rapid neutron capture^{4,21}. It has been pointed out^{19,20} that such a process can produce r-process nuclei in the observed mass range with abundance peaks corresponding to the neutron 'magic' numbers²². This is particularly interesting because these abundance peaks are observed to be quite narrow and to be distinct from the corresponding s-process abundance peaks. More conventional r-process models involving supernova explosions require fine tuning of temperature and neutron densities throughout many different dynamical events in order to produce such sharp peaks^{23,24}. We see, however, that neutron-star decompression may give rise to sharp peaks as a consequence of the intrinsic nuclear physics rather than having to rely on hydrodynamic concordance throughout different events.

Neutron capture is extremely rapid and throughout most of the decompression the nuclei remain very close to the neutron drip line. However, because the β -decay timescale is then

comparable to the expansion timescale, radioactivity causes significant heating, so that the temperature may be of the order of 1 MeV throughout most of the decompression phase and the actual neutron excess may be somewhat below that of the zero-temperature neutron drip line, as required by the location of the r-process peaks.

Furthermore, antineutrinos should be emitted at energies of up to a few MeV. The β -decay states are usually excited states with energies again of a few MeV, and rapidly emit γ -ray cascades as they decay to stable states.

In a close neutron-star binary, the larger neutron star accretes most of its companion over a dynamical timescale of $\sim 10^{-2}$ s, liberating energy at a rate of $\sim 10^{54}$ erg s⁻¹, and copious neutrino emission is expected. The rate of mass infall onto the primary is within an order of magnitude of that expected from spherically symmetric core collapse, so the neutrino spectrum which is buffered by energy-dependent interaction cross-sections, should resemble the (now measured) burst from a core collapse. Note that in the coalescing case the matter in the envelope is hotter than in the isolated core collapse and it is therefore easier for the neutrinos to diffuse. The neutrino diffusion timescale is of the order of 1 s, but because the distribution of infalling material is biased towards the orbital plane, neutrinos may escape out of the plane on a significantly shorter timescale. A total energy output of $\sim 10^{53}$ erg is expected from the accretion process, a sixth of which is due to electron neutrinos (unfortunately, such neutrinos cannot be detected with existing detectors beyond ~ 100 kpc). An energy of $\sim 10^{51}$ erg is expected from antineutrinos that result from the de-neutronization of the companion during its decompression.

The electron neutrinos from the accreting primary and the antineutrinos from the disrupted secondary can produce electron-positron pairs where they collide out of the orbital plane. The efficiency with which antineutrinos are converted is roughly²⁵

$$\varepsilon \approx 7 \times 10^{-3} \alpha f [(\sigma_{\nu\bar{\nu}} / 10^{-44} \text{ cm}^2) (\bar{E}_\nu / 10 \text{ MeV})^2] \times \left[\frac{(E_\nu / 10^{53} \text{ erg})}{(\bar{E}_\nu / 10 \text{ MeV})} \right] (R / 10^6 \text{ cm})^{-1} (\tau / 0.1 \text{ s})^{-1}$$

where $\sigma_{\nu\bar{\nu}}$ is the cross-section for $\nu\bar{\nu} \rightarrow e^+e^-$, E_ν is the total energy emitted in neutrinos, $\bar{E}_\nu$ is the average energy of a neutrino; R is the size of the region in which the neutrino flux is large and τ is the duration of the neutrino burst; α is a geometrical factor that depends on the geometry of the neutrinosphere and f is the filling factor of the system.

The total energy converted to e^+e^- pairs and, ultimately, the time profiles of the neutrino emission make this estimate of ε rather uncertain. The energy of the primary γ -ray cascades accompanying the β -decay antineutrinos should be within an order of magnitude of this total energy, that is, $\geq 10^{50}$ erg.

A comparable or larger γ -ray flux might be produced from pair production by $\nu\bar{\nu}$ pairs emerging from, say, opposite sides of the primary because, although the geometry is less favourable, the greater total energy and the greater compactness may more than compensate. Here the key uncertainty concerns the matter that presumably envelopes the more massive neutron star as it accretes onto it. Because this matter is intensely heated by its own radioactivity, we cannot draw too many parallels with previously catastrophic-mass-transfer models. In order that the γ -rays emerge before complete degradation, they must be produced beyond the accretion column, and this may require too small an intersection angle between the neutrino and antineutrino trajectories to sustain a centre-of-mass energy much above $2 m_e c^2$.

The γ -ray fireball has such a large compactness parameter that any soft photons produced within are Compton-upscattered and kinetic equilibrium is attained with the remaining pairs. The fireball is optically thick for temperatures $T > 10^9$ K, and e^+e^- pairs are produced. The fireball will expand until most of

the pairs have annihilated irreversibly, and it then becomes optically thin. Although the photons are redshifted during the expansion in the frame of the co-moving fluid, the bulk expansion at the onset of transparency causes a blueshift in the frame of the observer by the same factor. The overall spectrum will be a modified black-body spectrum with the same temperature as initially but with fewer hard photons²⁶. A large uncertainty in the end result of this process stems from the potential appearance of a thick baryonic wind. If a large amount of baryonic matter is mixed in with the fireball, most of the available energy will be used in accelerating it and the γ -ray signal will then not be so strong.

For a luminosity of 10^{47} erg s⁻¹ emitted from a region of size 2×10^7 cm, the temperature of the region is $\sim 10^9$ K. The spectrum and timescale should be similar to those of a typical γ -ray burst. As the Universe is transparent to soft γ -rays^{27,28}, it should be detectable to distances of ~ 100 Mpc with fluences typical of γ -ray bursts (10^4 eV cm⁻²), if the energy output is $\sim 10^{46}$ erg. However, we do not expect such radiation to have cyclotron absorption features, nor a significant non-thermal tail above several MeV. These two features are relatively common for γ -ray bursts, so that the scenario we propose can explain only a subset of them, that is, 'featureless' γ -ray bursts. Such featureless bursts would be extragalactic and thus isotropic. Statistics on featureless γ -ray bursts may thus provide an estimate of the rates of neutron-star coalescence events.

In numerical models of cataclysmic mass transfer, a small fraction ($\sim 0.3\%$) of the transferred mass is expelled to infinite distances. It is possible that radioactive heating, which can supply an energy of $\sim 3 \times 10^{-3} Mc^2$, can cause most of the disrupted star to explode and escape to infinity. There is $\sim 10^5 M_\odot$ per galaxy of material produced by the r-process, and a few times $10^3 M_\odot$ of the heavy r-process material that might be favoured in this type of event. If we assume that $\sim 0.3\%$ of the mass is ejected per event, then such events should occur at a rate of 3×10^5 to 3×10^7 times per galaxy per Hubble time (the lower limit applies if we wish to explain the heavy r-process material in this way and the upper limit applies if we wish to explain all r-processed material). These rates bracket the rate obtained by Clark *et al.*³ of 3×10^{-4} events per year per galaxy. If a subset of these events is also to be able to account for millisecond pulsars, then we require only about 10^{-5} events per year, or 10^5 per galaxy per Hubble time. Here we have assumed an average spin-down time for a typical millisecond pulsar of 10^9 years, and that there are currently about 10^4 millisecond pulsars in our Galaxy.

This scenario makes two simple observational predictions. First, assuming that $\sim 10^5$ galaxies are within 100 Mpc and that the bursts are indeed detectable out to that distance, then an occurrence of $\sim 10^{-4}$ per galaxy per year yields a detection rate of 10 per year. With the oriented scintillation spectrometer experiment on the Gamma Ray Observatory, it will be relatively straightforward to distinguish featureless, highly thermal γ -ray bursts from others. Should such a class be identified, we suggest that it would be worthwhile to check for identifications of such bursts with galaxies. Second, gravitational-radiation events of this nature should be detectable with a 30σ signal up to a distance of 100 Mpc and with a 3σ signal up to a distance of 1,000 Mpc by the proposed Caltech-MIT Gravitational Wave Detector². The rate of stronger events should be comparable to that of γ -ray bursts of this kind, and the coincidence of such γ -ray bursts with gravity waves may in fact provide the most stringent observational test of the scenario. Verification would imply that our model identifies the site of the astrophysical r-process. $\square$

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Observation of interfacial atomic steps during silicon oxidation

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SILICON oxidation is perhaps the most common process in the fabrication of electronic devices. The silicon/silicon dioxide interface has excellent electrical properties which are employed in field-effect devices and surface passivation, and are arguably responsible for the dominance of silicon technology in the micro-electronics industry. As device dimensions shrink, the need for high-quality oxides grown at increasingly lower thicknesses and temperatures has demanded ever more perfect, extremely flat Si/SiO₂ interfaces. Here we use transmission electron microscopy to image exposed and buried monatomic interfacial steps at the Si/SiO₂ interface during oxidation. The results show that extremely flat boundaries can be formed at low temperatures and that atomic steps do not play a dominant role in the oxidation process itself. A simple bulk-terminated Si/amorphous SiO₂ interface is consistent with our data.

The nature of the terminating Si lattice at the Si/SiO₂ boundary influences the transport of carriers in metal-oxide-semiconductor field-effect transistors (MOSFETs), and correlation has been observed between apparent interface roughness and Hall mobility¹. In the high-resolution transmission electron microscopy (HRTEM) study of ref. 1 it was emphasized that only statistical properties of interface roughness could be identified because of the implicit projection through ~ 100 Å specimen thickness. One detailed study² of Si/SiO₂ interfacial roughness with low-energy electron diffraction (LEED) assumed that the Si/SiO₂ interfacial structure was unperturbed by stripping of the oxide in HF. Other experiments³⁻⁷ which examined the stoichiometry of the SiO_x layer adjacent to the interface suggest an abruptly terminating Si lattice. In a recent HRTEM study of Si layers and their native oxides⁸ deposited by molecular beam

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